

SHASHWAT SAHU

Bangalore, India | +91 9900000000 | shashwatsahu2103@gmail.com | LinkedIn | GitHub | Portfolio

EDUCATION

Bhilai Institute of Technology (BIT) — Bachelor of Technology (B.Tech) in Robotics & Artificial Intelligence (2026) | **CGPA: 8.8 / 10**

KEY TECHNICAL PROJECTS

Shashwat's Bot: Custom 12-DOF Quadruped Robot [Raspberry Pi, ESP32, MPU6050 IMU, TB6612FNG, Python]

- Fully modeled mechanical quadruped chassis in Autodesk Fusion 360, optimizing leg geometries for 3D printing and structural modularity.
- Constructed hardware control architecture combining Raspberry Pi 4 high-level planner with ESP32 real-time motor control loops.
- Integrated MPU6050 IMU sensor feedback and implemented custom Python inverse kinematics gait algorithms for dynamic balance stabilization.

SynaptiArm-6X: Neural-Augmented 6-DOF Manipulator [Python, C++, Arduino, Machine Learning, Fusion 360]

- Engineered bio-electrical signal processing pipeline (Band-pass + Notch filtering, RMS feature extraction) achieving 94% intent accuracy at <30ms latency.
- Developed Arduino serial servo control loops for autonomous pick-and-place operation using 3D-printed 6-DOF inverse kinematics arm.

Finity — Premium AI Financial Companion [React 18, FastAPI, Three.js, Supabase, Sarvam AI]

- Designed production financial platform featuring context-aware Indian taxation advice, 50/30/20 budgeting engine, and Groq LLM integration.
- Built interactive investment simulator with probabilistic Monte Carlo mathematical visualizations using React 18, FastAPI, and Three.js.

RAVR — Hyperlocal Cultural Radar [Next.js 16, React 19, Supabase, Leaflet, Tailwind CSS]

- Engineered real-time urban event discovery platform for Bangalore using Next.js 16 App Router, Leaflet map views, and Supabase real-time subscriptions.

NyaySathi: Voice-First Legal Intelligence System [Python, FastAPI, Qdrant, Voice AI, RAG]

- Integrated Qdrant vector database semantic search and RAG retrieval pipeline over Indian legal statutes with voice-to-petition automated drafting.

Loan Genie: AI-Powered Loan Assessment [React 18, FastAPI, Three.js, Supabase, Groq LLM]

- Trained Random Forest Classifier for instant loan approval prediction and deployed 3D lip-synced digital avatar assistant in 11 Indian languages.

WORK & LEADERSHIP EXPERIENCE

AI & Robotics Operations Manager Intern — Incanto Dynamics Pvt. Ltd. (Jan 2026 – Apr 2026)

- Leading on-ground deployment, testing, and integration of industrial robotic systems and AI-driven operations within active factory environments.

Senior Product Manager — AIESEC in India (Feb 2025 – Aug 2025)

- Engineered conversion funnels and automated acquisition pipelines, scaling user acquisition workflows to directly generate ₹1.6L in value.
- Led a cross-functional team of 5 using agile frameworks, improving overall funnel efficiency by 35% and conversion rates by 40%.

Senior Marketing Manager — AIESEC in India (Aug 2025 – Jan 2026)

- Designed cohort-targeted growth campaigns and programmatic attribution models, boosting qualified lead acquisition by 60% across student networks.

Artificial Intelligence Intern — 1Stop.ai (Apr 2024 – Jun 2024)

- Trained, evaluated, and benchmarked machine learning models, integrating AI features into production workflows via RESTful API endpoints.

Club Operations Lead & Technical Manager — Robotics Cell, BIT (Jan 2024 – Jan 2025)

- Directed student robotics teams through physical bot build cycles, managed hackathon participations, and mentored 30+ juniors on embedded C/C++ control systems.

TECHNICAL SKILLS & COMPETENCIES

Programming: Python, C++, C, JavaScript

Robotics & Physical AI: ROS2, Arduino, Raspberry Pi, Gazebo, Fusion 360

AI & ML: PyTorch, TensorFlow, OpenCV, Scikit-learn

Tools & Frameworks: Git, FastAPI, React.js, Supabase